Beyond the Needle: Stress-Testing Long-Context Intelligence

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1 Tasks Performed

Our team has made substantial progress across all components of the LongBench diagnostic framework. We describe the completed work by module below.

1.1 Data Pipeline and Pilot Inference

The data pipeline has been fully implemented, covering all three main modules. The dataset loader handles all 503 instances of LongBench v2 (Bai et al., 2024), ensures that all records have the necessary fields (id, input, question, choices, answer, and category) present, and allows for categorizing according to the task type. The truncation module uses the Hugging Face tokenizer for processing the context field and implements a token budget of choice, which results in the creation of JSONL splits in two variants – 8k.jsonl and 32k.jsonl. Afterward, a stratified pilot inference script selects 50 examples from the entire data distribution equally from each of the six task categories using a consistent random seed, runs the examples using the Direct Answer prompt at 8K token budget using the LocalHFBackend, and outputs structured records following the shared result schema.

1.2 Model Backend and Evaluation Harness

The abstract model interface sets a standard LLMBackend contract. Each backend must return prediction, input_tokens, output_tokens, and latency_ms for every inference call. The Llama-3.1-8B-Instruct implementation uses the Hugging Face pipeline with temperature set to 0 for consistent results, and measures latency with perf_counter. The evaluation harness includes accuracy(), per_category_accuracy(), and bootstrap_ci(), using vectorized NumPy resampling over 10,000 iterations at a 95% confidence level. Unit tests check all three functions against known results..

1.3 Prompting Framework and E2A Pipeline

All prompt modules have been developed. The response parser uses extract_choice() with a regex that provides a unique answer letter or None if there is no unambiguous answer in the model output. The Direct Answer module creates prompts according to the uniform schema used in all the experiments. Two E2A modules have been created – one for evidence extraction, and another one for creating an answer based on extracted evidence. The orchestrator combines both modules and processes any LLMBackend; it adds up token counts for both requests and returns a result in the uniform schema.

1.4 Logging and Span Detection

The logging module defines an InferenceLogger class which is a context manager that ensures all eleven results attributes are fulfilled before adding the record to the log as a JSON line. The span detection module finds the sentence containing the answer in single-document questions using BM25Okapi (rank_bm25) and outputs the sentence along with its index number, normalized score, and whether the sentence is relevant or not. It throws ValueError for task types that are not for single document question answering where the perturbation method cannot be applied.

1.5 Annotation Schema and Inter-Annotator Agreement

The annotation schema defines three primary error categories: Retrieval Failure (RF), Reasoning Failure (RsF), and Instruction Non-Compliance (INC); together with a binary Hallucination (HALL) secondary attribute. The annotation guidelines include a flow chart for making decisions on the type of error, nine examples for each of the three error categories, three non-examples, instructions for marking hallucinations, and a procedure for adjudicating conflicts between annotators. The inter-rater reliability calculator uses scikit-learn to calculate Cohen's κ and reports all disagreements between annotators' annotations.

2 Potential Risks and Challenges

GPU resource constraints. The full experiments with the 64K and 128K contexts demand continuous GPU availability. The computations on these budget levels are not only time-consuming but also consume a large amount of memory; thus, competition for access to the GPUs may cause a delay in obtaining the results of the third phase. Failure to conduct 128K inference on Llama-3.1-8B-Instruct would leave the graphs incomplete.

Low inter-annotator agreement. Subjective decision-making must occur in determining whether a mistake was due to the recall process or reasoning process. In case the pilot Cohen's κ is lower than a desired level (for instance, $\kappa < 0.6$), then the annotations will not be reliable, hence rendering any error distribution analysis invalid.

Model output parsing failures. In case the model consistently outputs answers that lack a clear choice among A/B/C/D, the overall predictive performance decreases due to higher estimation variability. This becomes especially common when the length of context is relatively long because models usually output lengthy explanations before providing the actual answer.

Needle-position perturbation scope. Span detection using BM25 may not work for problems involving multiple documents or source code repositories if there is no localization of evidence into a particular sentence. It will be challenging to generate positional variants from a sufficient portion of LongBench v2 during the project timeline.

Integration and schema drift. Having five writers contributing towards the JSONL output results file increases the chances of errors such as omissions and type mismatches that will break the code that computes metrics and annotations later down the line, after the deadline for corrections has passed.

3 Risk Mitigation Plan

In an attempt to tackle the problem of GPU limitations, we will immediately schedule 64K and 128K jobs to gain priority in the cluster. If the 128K inference cannot be done in time, we will state our results for the lengths of 8K, 32K, and 64K explicitly and indicate that 128K inference was not done as a contingency plan.

In case the pilot kappa score is less than $\kappa = 0.6$, a calibration exercise involving category pairs that have been most often confused will precede actual coding. In addition, the decision flowchart used in the coding manual will be modified to include more decision triggers, and there will be a third coder to break ties.

The None-prediction percentage will be tracked during the pilot experiments. If it breaches 10%, regardless of the token limit, Meghna will increase the formatting constraint in the Direct and E2A prompts and repeat the experiment with that subset. Predictions that have been classified as 'None' will be disregarded when calculating accuracy but flagged as potential instruction non-compliance cases.

The study involving needle positioning perturbation will only include single document QA cases where BM25 confidence score is above 0.5 to reduce noise due to span selection uncertainty. In case the subset created is not sufficiently large for statistical analyses, the needle positioning perturbation will be presented as exploratory evidence while main conclusions will be based on the context length scaling study.

Divyam's InferenceLogger is already validating mandatory fields when data is written. Our solution will include deploying a common validation utility, which is to be used by all collaborators prior to committing their results files, thereby making sure that schema non-compliant files do not enter our metrics or annotation pipeline.

The subsequent stages of implementation will proceed in accordance with the plan developed at the initial stage of the project. Stage 3 (conducted full experiments on 64K and 128K) will be carried out right after the writing of this paper. Stage 4 (full annotation and inter-rater agreement) will be done simultaneously as data is received.

4 Individual Contributions

Ansh Bhatt: I was responsible for the data pipeline, handling everything from loading and validating the raw LongBench v2 dataset to truncating contexts to fixed token budgets and writing the cleaned splits to disk. I also ran the initial pilot inference at the 8K budget, which produced the first round of results the rest of the team used for annotation and evaluation.

Shlok Bhura: I built the model backend infrastructure and the evaluation harness. This included defining the shared interface all model backends must follow, integrating Llama-3.1-8B-Instruct for local inference, and implementing the accuracy and confidence interval computation that the team uses as the authoritative metrics pipeline.

Rudrani Chavarkar: I led the annotation effort, designing the error taxonomy that classifies model failures into retrieval, reasoning, and instruction non-compliance categories. I created the annotation guidelines that annotators follow, coordinated the pilot labeling round, and built the inter-annotator agreement pipeline to measure labeling consistency across the team.

Meghna Nair: I owned the prompting layer of the project, designing both the Direct Answer and Extract-Then-Answer strategies and building the full two-stage E2A inference pipeline. I also ran the pilot experiments at the 32K budget, giving the team its first comparison point between context lengths and prompting approaches.

Divyam Shah: I handled observability and span analysis for the project. I built the structured logging system that records every inference call with full metadata, ensuring all downstream analysis has reliable, consistently formatted data to work from. I also developed the component that automatically identifies answer-relevant passages in the context, which underpins the needle-position perturbation experiment.

References

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